



JSS 2 MATHEMATICS LESSON NOTES (THIRD TERM)

WEEK 5: BEARINGS AND DISTANCES II

TOPIC: Constructing Triangles; Bisecting Angles

A. TRIANGLE CONSTRUCTION

Definition:

Triangle construction is the accurate drawing of a triangle using given measurements (sides and/or angles) with instruments such as a **ruler, compass, and protractor**.

BASIC RULE (VERY IMPORTANT):

A triangle can only exist if:

$$A + b > c$$

Meaning: the sum of any two sides must be greater than the third side.

B. TYPES OF TRIANGLE CONSTRUCTION

1. SSS (Three Sides Given)

Example 1:



Construct triangle ABC:

- $AB = 7 \text{ cm}$
- $BC = 6 \text{ cm}$
- $AC = 5 \text{ cm}$

Step-by-step method:

1. Draw base $AB = 7 \text{ cm}$ using ruler.
2. Set compass radius = 5 cm (AC), draw arc from A.
3. Set compass radius = 6 cm (BC), draw arc from B.
4. Intersection of arcs = point C.
5. Join AC and BC .

Result:

Triangle ABC is completed accurately.

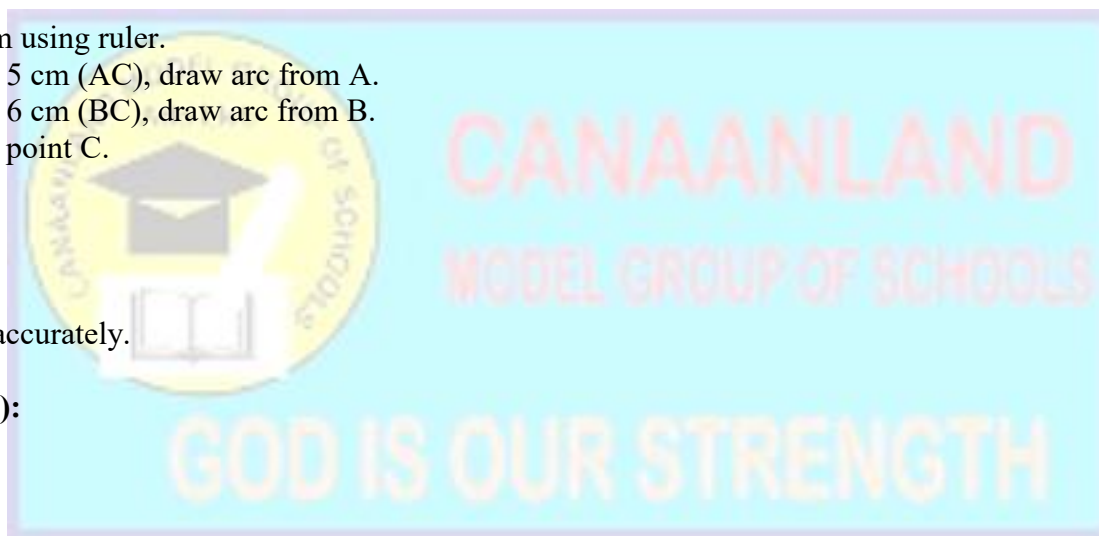
Example 2 (More Complex):

Construct triangle PQR:

- $PQ = 8 \text{ cm}$
- $QR = 6 \text{ cm}$
- $PR = 4 \text{ cm}$

Solution Steps:

1. Draw $PQ = 8 \text{ cm}$





2. Arc from P radius 4 cm
3. Arc from Q radius 6 cm
4. Intersection = R
5. Join PR and QR

Example 3 (INVALID CASE):

Can you construct 3 cm, 4 cm, 8 cm triangle?

Check:

$3 + 4 = 7$ ✗ (not greater than 8)

✓ Conclusion: Triangle cannot be formed.

C. SAS (Two Sides and Included Angle)

Example 4:

Construct triangle ABC:

- $AB = 6$ cm
- $AC = 5$ cm
- $\angle BAC = 50^\circ$

Steps:

1. Draw $AB = 6$ cm
2. At A, construct 50° using protractor
3. On that line, measure $AC = 5$ cm





4. Join C to B

Example 5 (Real Application Style):

Construct:

- $XY = 7 \text{ cm}$
- $YZ = 5 \text{ cm}$
- $\angle XYZ = 65^\circ$

Steps:

1. Draw XY
2. Construct 65° at Y
3. Mark $YZ = 5 \text{ cm}$
4. Join XZ

D. ASA (Two Angles and One Side)

Example 6:

- $AB = 7 \text{ cm}$
- $\angle A = 60^\circ$
- $\angle B = 50^\circ$

Steps:

1. Draw $AB = 7 \text{ cm}$
2. At A construct 60°





3. At B construct 50°
4. Lines intersect at C

Key Idea:

$$\angle A + \angle B + \angle C = 180^\circ$$

So:

$$\angle C = 180^\circ - (60^\circ + 50^\circ) = 70^\circ$$

E. SPECIAL TRIANGLES

1. Equilateral Triangle

- All sides equal
- All angles = 60°

Example: side = 6 cm

→ Construct using equal arcs from endpoints

2. Isosceles Triangle

- Two equal sides
- Two equal angles

3. Right-Angled Triangle

Example 7:





Base = 5 cm, height = 4 cm

Steps:

1. Draw base
2. Construct 90° at one endpoint
3. Mark height = 4 cm
4. Join hypotenuse

F. ANGLE BISECTION

Definition:

Angle bisection is dividing an angle into TWO equal parts using compass.

Example 8:

Bisect 80°

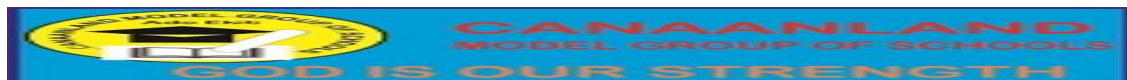
Steps:

1. Draw 80° angle
2. Draw arc from vertex cutting both arms
3. From both points, draw intersecting arcs
4. Join vertex to intersection

Result:

$$40^\circ + 40^\circ = 80^\circ$$





Example 9:

Bisect 120°

Result:

$60^\circ + 60^\circ$

Example 10:

Bisect $90^\circ \rightarrow 45^\circ + 45^\circ$

G. CONSTRUCTING COMMON ANGLES USING BISECTION

1. 45°

Construct $90^\circ \rightarrow$ bisect

2. 30°

Construct $60^\circ \rightarrow$ bisect

3. 15°

Bisect 30°

4. 22.5°

Bisect 45°





H. PERPENDICULAR BISECTOR

Definition:

A line that divides another line into two equal halves at 90° .

Example 11:

Line AB = 8 cm

Steps:

1. Draw AB
2. Draw arc from A and B (same radius)
3. Intersect arcs above and below
4. Join intersections

Result:

- Cuts AB exactly in half
- Forms 90° angle



I. APPLICATION IN BEARINGS

Example 12:

From point P:

- A at 070° , 5 cm



- B at 145° , 6 cm

Steps:

1. Draw North line
2. Plot A and B using compass
3. Join A to B
4. Measure bearing

J. IMPORTANT EXAM TIPS

- ✓ Always use sharp pencil
- ✓ Always label points clearly
- ✓ Check triangle validity first
- ✓ Ensure compass radius is constant
- ✓ Verify angle sum = 180°

(GUIDED PRACTICE)

- ✓ SSS construction
- ✓ SAS construction
- ✓ ASA construction
- ✓ Angle bisection
- ✓ Line bisection
- ✓ Mixed geometry reasoning

